

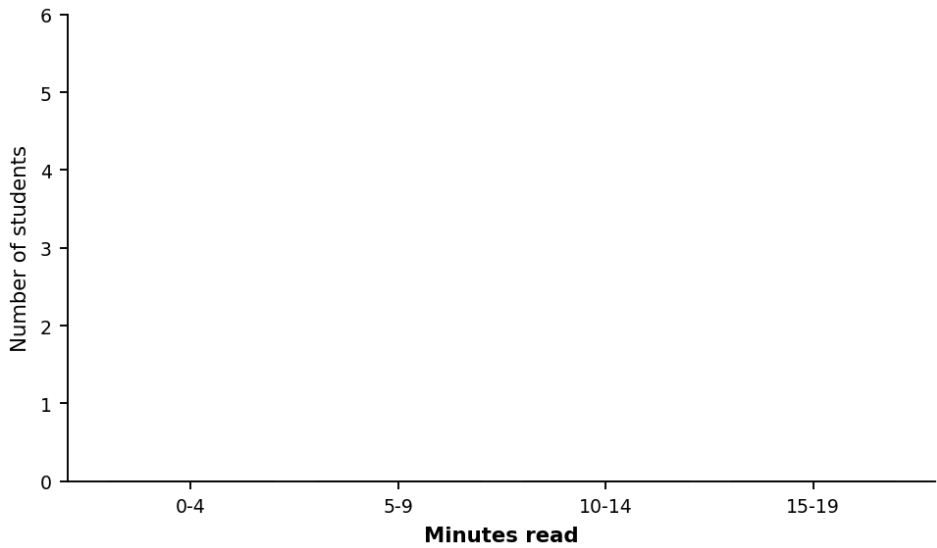
Test 8 — Question Paper

**38 items.** Suggested time: 95 minutes (about 2 minutes 30 seconds per item).  
A four-function calculator may be used **only** on items marked *calculator*.  
The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.  
Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. A data set is shown.

0, 2, 5, 6, 8, 9, 11, 12, 15, 17, 19, 19

Build the histogram by setting the height of each bar.



Mark your answer on the diagram above.

Show your work

graphing (grid)

2. A pond loses 4 centimeters of water each week during a dry spell.

Which integer represents the total change in water level after 4 weeks? Enter your answer.

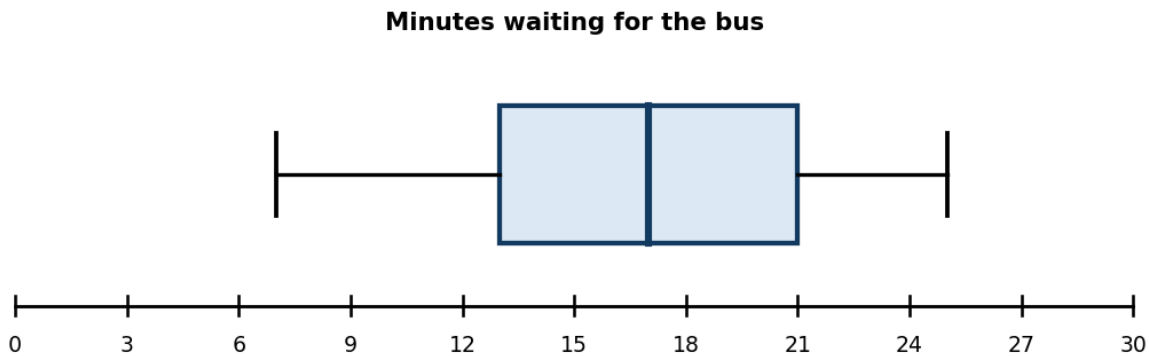
Answer: \_\_\_\_\_

Show your work

calculator · equation editor

3. The box plot shows a set of data.

What is the maximum? Enter your answer.



Answer: \_\_\_\_\_

Show your work

equation editor

4. Point A is located at  $(-2, -2)$ . Point A is reflected across the y-axis.

What are the coordinates of the reflected point?

- A.  $(2, -2)$
- B.  $(2, 2)$
- C.  $(-2, -2)$
- D.  $(4, -2)$

Show your work

multiple choice

5. Four sensors on a reef near Key Biscayne report these elevations, in feet relative to sea level:

$-47$ ,  $-34$ ,  $-25$ ,  $-12$

Which elevation is farthest from sea level?

- A.  $-12$
- B.  $-47$
- C.  $-34$
- D.  $-25$

Show your work

calculator · multiple choice

6. The table shows equivalent ratios of juice to water.

Enter the number that belongs in the box marked ?

| Cups of juice | Cups of water |
|---------------|---------------|
| 4             | 7             |
| 16            | ?             |

Answer: \_\_\_\_\_

Show your work

equation editor

7. Enter the prime factorization of 84. Use ^ for exponents and × between factors, for example 2^2 × 3.

Answer: \_\_\_\_\_

Show your work

calculator · equation editor

8. Which value of x makes the inequality  $x > 14$  true?

- A. 15
- B. 14
- C. 12
- D. 8

Show your work

multiple choice

9. What is the greatest common factor you would use to rewrite  $18 + 54$  as a product? Enter your answer.

Answer: \_\_\_\_\_

Show your work

calculator · equation editor

**10.** What is the product of  $\frac{3}{5}$  and  $\frac{3}{8}$ ?

- A.  $\frac{6}{13}$
- B.  $\frac{9}{40}$
- C.  $\frac{39}{40}$
- D.  $1\frac{3}{5}$

Show your work

calculator · multiple choice

**11.** At 6 a.m. the temperature was  $-5^{\circ}\text{F}$  in a walk-in freezer. It rose  $20^{\circ}\text{F}$  and then fell  $15^{\circ}\text{F}$ .

What is the temperature now, in  $^{\circ}\text{F}$ ? Enter your answer.

Answer: \_\_\_\_\_

Show your work

calculator · equation editor

**12.** A recipe uses 32 cups of rice and 20 cups of water.

Write the ratio of rice to water in simplest form. Use the form a:b.

Answer: \_\_\_\_\_

Show your work

equation editor

**13.** What is the value of  $|-8\frac{7}{10}|$ ?

- A. 8.7
- B. -0.11
- C. 17.4
- D. -8.7

Show your work

calculator · multiple choice

**14.** Marcos buys 5 cases of water at \$2.63 each and pays a \$4.02 delivery fee.

What is the total cost?

- A. \$33.25
- B. \$13.15
- C. \$17.17
- D. \$22.73

Show your work

calculator · multiple choice

**15.** Which expression is equivalent to  $7x + 12 + 8x$ ?

- A.  $15x + 12x$
- B.  $15x + 12$
- C.  $56x + 12$
- D.  $27x$

Show your work

multiple choice

**16.** Elena shared 30 guavas equally among 5 baskets.

Which equation can be used to find  $x$ , the number of guavas in each basket?

- A.  $30x = 5$
- B.  $x + 5 = 30$
- C.  $x \div 5 = 30$
- D.  $5x = 30$

Show your work

multiple choice

**17.** Write 0.75 as a percent. Enter your answer without the % symbol.

Answer: \_\_\_\_\_

Show your work

calculator · equation editor

**18.** A teacher has 18 pencils and 30 erasers to put into identical bags with nothing left over.

What is the greatest number of bags she can make?

Answer: \_\_\_\_\_

Show your work

calculator · equation editor

**19.** A pair of headphones costs \$45.00 and is on sale for 50% off.

What is the sale price, in dollars? Enter your answer.

Answer: \_\_\_\_\_

Show your work

equation editor

**20.** Which question is **not** a statistical question?

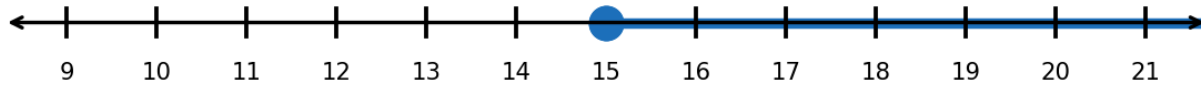
- A. How many books did each student in the class read last month?
- B. How many text messages did each student send yesterday?
- C. How old is my teacher?
- D. What are the heights, in inches, of the students in the band?

Show your work

multiple choice

**21.** The number line shows the solution to an inequality.

Which inequality does the graph represent?



- A.  $x \geq 15$
- B.  $x \leq 15$
- C.  $x > 15$
- D.  $x < 15$

Show your work

multiple choice

**22.** A box shaped like a right rectangular prism has a volume of 150 cubic inches. Its base has an area of 25 square inches.

What is the height of the box, in inches?

- A. 25
- B. 125
- C. 6
- D. 12

Show your work

multiple choice

**23.** Nico recorded these four numbers in her math journal: -2.5,  $-\frac{3}{4}$ ,  $\frac{3}{5}$ , 2.

Which number is the least?

- A. -2.5
- B.  $\frac{3}{5}$
- C.  $-\frac{3}{4}$
- D. 2

Show your work

calculator · multiple choice

**24.** A fence is 6 yards long.

How many feet long is the fence? Enter your answer. (1 yard = 3 feet)

Answer: \_\_\_\_\_

Show your work

equation editor

**25.** A data set is shown.

8, 17, 21, 23, 24, 27, 29

The value 8 is removed. What is the median of the new data set? Enter your answer.

Answer: \_\_\_\_\_

Show your work

equation editor

**26.** What is the greatest common factor of 48 and 72?

- A. 24
- B. 26
- C. 48
- D. 144

Show your work

calculator · multiple choice

**27.** What is the value of  $0.3^2$ ? Enter your answer.

Answer: \_\_\_\_\_

Show your work

calculator · equation editor

**28.** A diver off South Beach descends 19 feet below the surface, then rises 12 feet. Sea level is represented by 0.

Which integer represents the diver's position now?

- A. -7
- B. -31
- C. 7
- D. 31

Show your work

calculator · multiple choice

**29.** What is the product of 27.5 and 0.31?

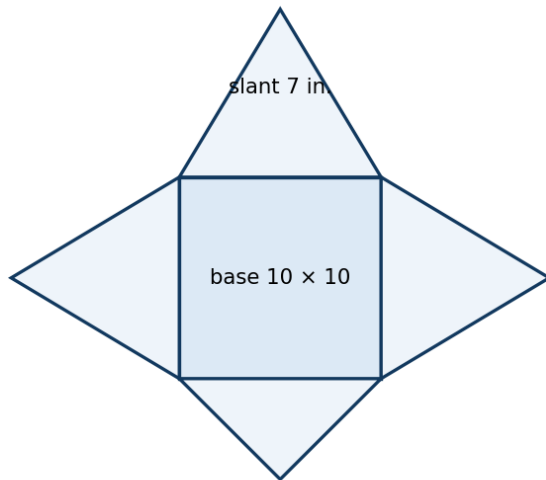
- A. 8.525
- B. 0.8525
- C. 852.5
- D. 85.25

Show your work

calculator · multiple choice

**30.** The net of a right rectangular pyramid with a square base is shown.

What is the surface area of the pyramid, in square inches?



- A. 380
- B. 240
- C. 100
- D. 280

Show your work

multiple choice

**31.** A triangle has an area of 15 square feet and a height of 3 feet.

What is the length of its base, in feet?

- A. 12
- B. 5
- C. 10
- D. 30

Show your work

multiple choice

**32.** Isabella had some stickers and gave away 3 of them. Now Isabella has 26 stickers.

Which equation can be used to find  $x$ , the number of stickers Isabella started with?

- A.  $x + 3 = 26$
- B.  $x - 3 = 26$
- C.  $3 - x = 26$
- D.  $x + 26 = 3$

Show your work

multiple choice

**33.** A pack of 8 bottles of water costs \$38.08.

What is the cost per bottle, in dollars? Enter your answer.

Answer: \_\_\_\_\_

Show your work

equation editor

**34.** What is the value of  $(2x + 8) \div 2$  when  $x = 8$ ?

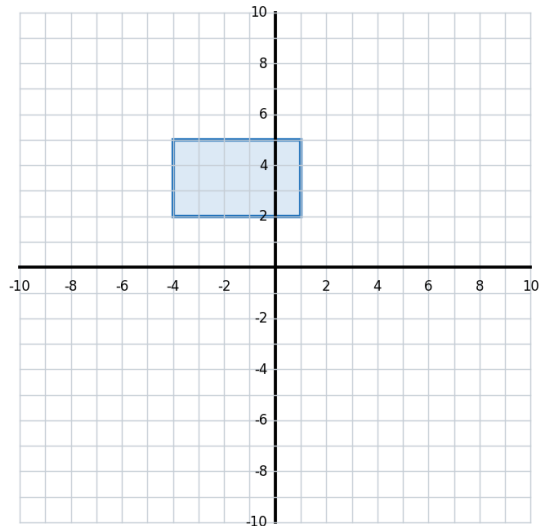
- A. 24
- B. 12
- C. 16
- D. 20

Show your work

multiple choice

**35.** A rectangle has vertices at  $(-4, 2)$ ,  $(1, 2)$ ,  $(1, 5)$  and  $(-4, 5)$ .

What is the area, in square units, of the rectangle?



- A. 8
- B. 16
- C. 15
- D. 30

Show your work

multiple choice

**36.** Ximena wrote the expression  $x/6$ .

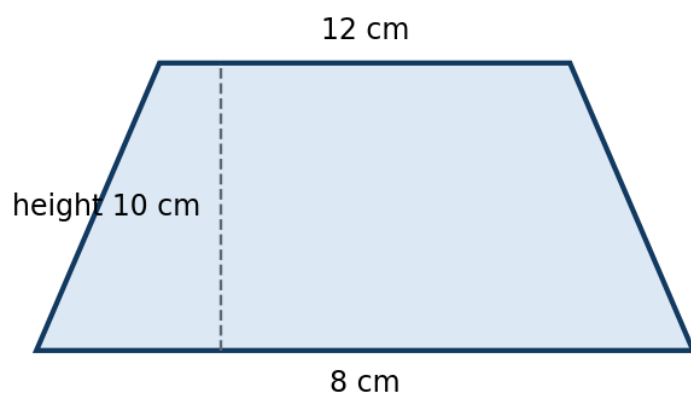
Which written description matches the expression?

- A. the quotient of a number  $x$  and 6
- B. 6 less than a number  $x$
- C. a number  $x$  divided by 6
- D. the product of a number  $x$  and 6

Show your work

multiple choice

37. What is the area of the trapezoid shown, in square centimeters?



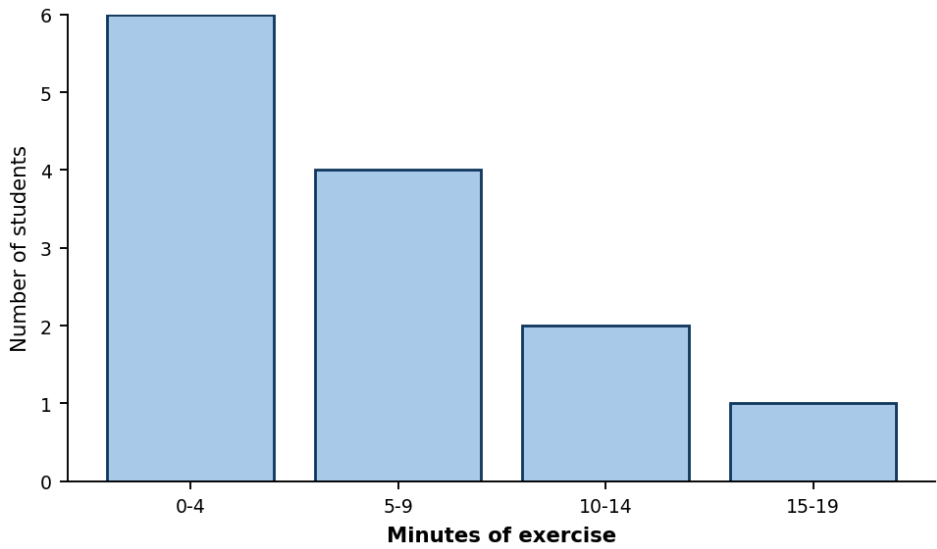
- A. 80
- B. 200
- C. 100
- D. 15

Show your work

multiple choice

38. The histogram shows the data collected from a group of students.

Which statement best describes the shape of the data?



- A. The data are skewed to the left, with most values in the higher intervals.
- B. The data are skewed to the right, with most values in the lower intervals.
- C. The data are symmetric, with about the same number of values on each side.
- D. The data have no values in the lowest interval.

Show your work

multiple choice

## Grade 6 Mathematics Reference Sheet

### Customary Conversions

|                      |                        |
|----------------------|------------------------|
| 1 foot = 12 inches   | 1 cup = 8 fluid ounces |
| 1 yard = 3 feet      | 1 pint = 2 cups        |
| 1 mile = 5,280 feet  | 1 quart = 2 pints      |
| 1 mile = 1,760 yards | 1 gallon = 4 quarts    |
| 1 pound = 16 ounces  | 1 ton = 2,000 pounds   |

### Metric Conversions

|                             |                             |
|-----------------------------|-----------------------------|
| 1 meter = 100 centimeters   | 1 liter = 1,000 milliliters |
| 1 meter = 1,000 millimeters | 1 gram = 1,000 milligrams   |
| 1 kilometer = 1,000 meters  | 1 kilogram = 1,000 grams    |

### Time Conversions

|                       |                   |
|-----------------------|-------------------|
| 1 minute = 60 seconds | 1 week = 7 days   |
| 1 hour = 60 minutes   | 1 year = 365 days |
| 1 day = 24 hours      | 1 year = 52 weeks |

### Formula

|                                         |                                  |
|-----------------------------------------|----------------------------------|
| Rectangular prism                       | $V = lwh$ or $V = Bh$            |
| $l$ = length, $w$ = width, $h$ = height | $B$ = area of base, $V$ = volume |